Contact

8625955351 (Mobile)
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Top Skills

AI Agents
GitHub
Linux

Languages

English (Full Professional)
Marathi (Native or Bilingual)
Hindi (Native or Bilingual)

Certifications

PCB design for Everyone using Autodesk Eagle

Honors-Awards

1st prize at ISTE Approved State level Project Exhibition TechfestC 2k25

3rd prize at Technovation, 2025 organized by IEEE SB, SSGMCE Shegaon

1st Prize at National level Event TECHTANTRA 2k25 in Tech-ExPO

1st prize in SPARK 2K25 National level Project Competition

2nd prize at district level Aavishkar 2k25 in Pure Science Category

Publications

The Future of Drones in Disaster Management: A New Era of Emergency Response

Innovative Techniques for Efficiency Enhancement and Noise Suppression in an Inverter

Patents

Hrishikesh Kakade

Firmware - Robotics | Space tech Enthusiast | intern @airbotix technology | Ex-intern @ideaForge | Ex-intern @adolfsolutions
Amravati, Maharashtra, India

Summary

Electronics & Telecommunication Engineering student with hands-on industry experience in UAV swarm firmware, RTK GPS setup, Li-ion battery development, PCB design, and IoT embedded systems. Published patent holder in UAV-based flood mapping. Proficient in C/C++, Python, Ai agents, STM32, ESP32, ROS2 Humble, and Pixhawk-based flight systems. Passionate about building reliable embedded and drone solutions for real-world impact.

Experience

Airbotix Technology
Embedded Firmware Intern
July 2026 - Present (4 months)
Gurugram, Haryana, India

As a Firmware development intern in R&D I am working with variety of domains like Firmware, Ai Agents, Flight Controller & GCS, Swarm & Precise Navigation, version control & certification.

- Firmware & Connectivity: Customized open-source embedded firmware using ESP-IDF for ESP modules, optimizing reliable 2.4GHz and 5GHz network communications.
- AI-Assisted Development & CLI: Leveraged AI agents (Gemini, Claude, DeepSeek, Antigravity, Opencode) and Command Line Interfaces (CLI) to accelerate coding, debugging, and Linux-based workflows.
- Flight Control & GCS: Integrated and configured Pixhawk Cube Orange flight controllers, utilizing Mission Planner and QGroundControl (QGC) for mission execution and understanding it for software development.
- Swarm & Precision Navigation: Worked with drone swarm operations using Skybrush and integrated Real-Time Kinematic (RTK) setups for centimeter-level positioning accuracy.
- Development & Compliance: Leveraged Linux, open-source tools, and Git/GitHub for version control, while managing technical documentation to secure DGCA product certification.

SYSTEM FOR MAPPING OF FLOOD AND EARLY WARNING CONFIGURATION BY MEANS OF AN UAV

Fiverr
Freelance
June 2026 - June 2026 (1 month)
Pune City, Maharashtra, India

During this period I have worked on ROS projects for clients.

- Developed ROS-based virtual geofencing to constrain robot navigation and analyzed computational graphs to optimize system communication.

VIHANG DRONE CLUB
2 years 10 months

Research Associate
August 2024 - May 2026 (1 year 10 months)

As a reasearch associate I have developed a Patented project like "System for Mapping of Flood and Early Warning Configuration by means of an UAV". Worked on Pixhawk, STM32 and written the reasearch paper "The Future of drones in Disaster Management: A New Era of Emergence Response"

Member
August 2023 - August 2024 (1 year 1 month)
Shegaon, Maharashtra, India

As a member I have designed a Club website, Obstacle avoidance car, learn about Drone fundamentals, Flight controllers and drone building and support in projects like RC plane, etc.

ideaForge
Intern
November 2025 - March 2026 (5 months)
Navi Mumbai, Maharashtra, India

As a Process intern I have streamlined UAV manufacturing processes during the early phases of the Zolt tactical platform, successfully bridging R&D with production to reduce cycle times and resolve on-site subsystem issues.

- Bridged R&D and Production: Closed the operational gap between research and manufacturing by conducting technical workshops, ensuring seamless handoffs and aligning cross-functional teams.
- Optimized Manufacturing Workflows: Developed comprehensive flow diagrams and cycle time charts on the Zolt tactical platform to analyze bottlenecks and streamline production phases.

- Reduced Cycle Times: Actively identified flaws in early product design and implemented process optimizations that directly reduced overall manufacturing cycle time.
- Spearheaded On-Site Troubleshooting: Resolved critical developmental and assembly issues on the floor for complex subsystems, including UAV batteries, Ground Control Station (GCS) assemblies, and composite materials.

Adolf Solutions
Trainee Software Development Engineer
June 2025 - July 2025 (2 months)
Pune District, Maharashtra, India

As a trainee software development engineer during my internship in R&D I have work on IoT projects and developed a crud software.

• Engineered a real-time IoT communication system utilizing two ESP microcontrollers.
• Scaled device connectivity from a shared local network to cross-network internet communication.
• Developed an independent web server to enable remote web access and real-time hardware triggering.
• Developed a custom web portal to automate critical business workflows including billing, inventory, and product management.
• Streamlined daily operations and eliminated manual data entry bottlenecks for the sales administration and production teams.
• Enhanced cross-functional efficiency by centralizing essential operational processes into a single, unified platform.

Education

Shri Sant Gajanan Maharaj College of Engineering, Shegaon
Bachelor of Engineering - BE, Electronics and telecommunication · (November 2022 - June 2026)

Shivaji Junior College and Highschool Nandgaon (Kh)
HSC, Science · (June 2021 - February 2022)

Shivaji Junior College and Highschool Nandgaon (Kh)
SSC · (June 2018 - March 2019)